Calcurion Maths Worksheets

GCSE Mathematics

Topic: Probability

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. A fair coin is tossed. Find the probability of heads. **[1]**

2. A fair six-sided dice is rolled. Find the probability of scoring a 4. **[1]**

3. A fair six-sided dice is rolled. Find the probability of scoring an even number. **[2]**

4. A bag contains 3 red and 5 blue balls. One ball is chosen at random. Find the probability it is red. **[2]**

5. A bag contains 4 red, 3 green and 5 blue balls. Find the probability of choosing a green ball. **[2]**

6. The probability of rain is 0.3. Find the probability it does not rain. **[1]**

7. Two fair coins are tossed. Find the probability of two heads. **[2]**

8. A fair dice is rolled. Find the probability of scoring a number greater than 4. **[2]**

9. A spinner has 5 equal sections numbered 1 to 5. Find the probability of scoring an odd number. **[2]**

10. Events A and B are mutually exclusive. $P(A) = 0.4$, $P(B) = 0.25$. Find $P(A \text{ or } B)$. **[2]**

11. Events A and B are independent. $P(A) = 0.3$, $P(B) = 0.5$. Find $P(A \text{ and } B)$. **[2]**

12. A card is drawn from a standard pack of 52. Find the probability it is a heart. **[2]**

13. A card is drawn from a standard pack. Find the probability it is an ace. **[2]**

14. The probability of passing a test is 0.8. Find the probability of failing. **[1]**

15. Two fair dice are rolled. Find the probability of scoring a total of 7. **[3]**

Section B – Hard (Q16–25)

16. A bag contains 5 red and 3 blue balls. Two balls are drawn without replacement. Find the probability both are red. **[4]**

17. A bag contains 4 red and 6 blue balls. Two balls are drawn with replacement. Find the probability both are blue. **[3]**

18. Two fair dice are rolled. Find the probability of scoring a double. **[3]**

19. A card is drawn from a pack. Find the probability it is a red king. **[2]**

20. Events A and B are independent. $P(A) = 0.4$, $P(A \text{ and } B) = 0.12$. Find $P(B)$. **[3]**

21. A tree diagram shows $P(\text{rain}) = 0.3$. If it rains, $P(\text{late}) = 0.6$; if not, $P(\text{late}) = 0.2$. Find the probability of being late. **[4]**

22. Two events are mutually exclusive. $P(A) = 0.35$, $P(B) = 0.45$. Find $P(\text{neither})$. **[3]**

23. A fair coin is tossed three times. Find the probability of exactly two heads. **[4]**

24. A bag contains n red and 5 blue balls. The probability of choosing a red ball is $\frac{2}{5}$. Find n. **[3]**

25. Two cards are drawn without replacement from a pack. Find the probability both are aces. **[4]**

End of Worksheet • *Show all working clearly.*